

Midterm Progress Report

An Open Project in Deep Learning: Hand Gesture Recognition System

TAM, Ka Ho; MAN, Tze Wa

1155175983@link.cuhk.edu.hk, 1155174636@link.cuhk.edu.hk

Advisor: Dr. LAM, King Tin

The Chinese University of Hong Kong, Sha Tin, Hong Kong
Bachelors of Engineering in Artificial Intelligence: System and Technology

1 Introduction

Hand Gesture Recognition (HGR) has played a vital role in driving fields such as virtual/augmented/mixed reality and human-machine interactions. In particular, HGR is frequently used to extract instructions or language conveyed through hand gestures. However, the task of gesture recognition typically involves a number of challenges, including the possible variability in the dynamics of gestures, change in environment, occlusion, and the scarcity of high-quality well-annotated data.

Rapid advancements in deep learning models and sophistication in easy-to-use frameworks for keypoint extraction in the fields of computer vision and gesture recognition, such as MediaPipe and DWPose, have driven the expectations and needs in gesture recognition systems that is capable of not only recognizing limited sets of static gestures, but also distinguishing within a large variety of dynamic gestures often with subtle differences.

Previous HGR frameworks tackle the problems of performance degradation under a change of the environment, including the background and the signer, by integrating a multi-modal approach, specifically, by including keypoints as one of the streams of inputs passed into the models [1] [2].

However, previous HGR frameworks mostly work with a fixed set of gestures, which may not be sufficient in the case that custom gestures may be required for some purposes like customized gesture control for games or computer interfaces, or in the case that obtaining the required full set of high-quality well-annotated gesture data is too costly.

While acquiring unannotated samples for training is far more feasible and less resource-intensive, Semi-Supervised Learning, which leverages a limited set of annotated data alongside a larger set of unannotated samples, emerges as a promising strategy.

In this project, the primary objective would be to develop a method that maintains effectiveness and accuracy, even in scenarios where only a limited number of reference samples is available. This project aims to explore the possibilities that semi-supervised and contrastive learning approaches could enable the models to learn a more structured and general representation of various hand gestures.

2 Related Work

2.1 Image Recognition

EfficientNet [3] has been an efficient and accurate benchmark for image recognition tasks, making it highly effective as the backbone for many downstream fine-tuning tasks. In this project, we will adopt EfficientNet to process the image streams of our models.

2.2 Contrastive Learning Approaches

In static image recognition, contrastive learning techniques, in particular, Siamese Networks [4], has played a crucial role under unsupervised and semi-supervised settings by utilizing shared parameters among the encoders for contrastive pairs. The architecture has contributed to the development of later frameworks such as SimCLR [5], MOCO [6], BYOL [7], etc., and have shown success on utilizing large sets of unannotated data by generating positive samples via data augmentation.

2.3 Skeleton-Based Approaches

With sophisticated pre-trained pipelines for extracting keypoints of hand and body gestures from videos, such as DWPose [8] and MediaPipe [9], many "skeleton-based" HGR models, like Sung et al. [10], TwoStreamSLR [1], and STGCN-GR [11], are integrating keypoints as a new input stream. This integration can theoretically improve robustness against environmental variations, given a reliable pre-trained keypoint extraction model like MediaPipe. Additionally, a wider variety of augmentations, such as translation and rotation of individual joints, are easier to implement compared to using the video stream directly. In the first half of this project, we will also be exploring the potential of using contrastive learning over the keypoints.

2.4 Dynamic Gesture Recognition

While HGR models for static gesture recognition have become sophisticated, dynamic gesture recognition has remained a challenging task, especially over a large number of classes. Methods like TwoStreamNetwork [1] and SlowFastSign [12] perform convolution also across the time dimension to integrate temporal data and has still been in the leading position over the task of sign language recognition.

2.4.1 With Contrastive Learning Approaches

Techniques such as Contrastive Predictive Coding [13] and Temporal Contrastive Learning [14], e2eET [2] has been discussed in the literature.

2.4.2 With Skeleton-Based Approaches

Methods such as STGCN [15], STGCN-GR [11] have been discussed in the literature.

However, due to the difficulties in exploring methodologies to integrate the temporal dimension with our methods, we would leave this part to the second half of this project if the first half goes smoothly.

3 Project Goal

While most common gesture recognition models focuses on a fixed set of gestures, this project aims to improve the flexibility and usability of gesture recognition systems, addressing the needs of custom gestures in some applications and ideally reducing the time and resources required for fine-tuning.

In the process, we aim to explore the possibilities that semi-supervised and contrastive learning approaches could enable the models to learn a more structured and general representation of various hand gestures.

3.1 Scenarios of Application

We aim to enable quick adaptation of the system to tasks like game control, screen control (zooming, etc), and sign language (such as gloss extraction based on similarity in gestures), with minimal re-training / finetuning if possible.

3.2 Secondary Goals

- Enable custom gesture addition without largely retraining the network.
- Investigates if contrastive learning works better than transfer learning on embedding model pre-trained via supervised learning methods in terms of accuracy and efficiency.
- Investigates if our approach could possibly utilize a large labelled dataset for contrastive learning to minimize fine-tuning efforts given a relatively small gesture dataset.
- Match and annotate gestures from videos.

4 Methodology

The preliminary model utilizes the video stream as input, with input size of 512×512 . The model consists of a backbone model, in which case the EfficientNetB7 model pretrained on ImageNet is adopted [3]. We removed the Multi-Layer Perceptron (MLP) classifier head from the backbone and replaced it with a customized 3-layer MLP, with Batch Normalization, ReLU activation and a Dropout Layer after each linear layer except the last.

Currently, we have managed to perform tests on two versions of the model, one trained with Cross Entropy (CE) loss and one trained with the Siamese Architecture with vanilla triplet loss.

5 Datasets

In this project, we plan to use two groups of datasets, one for static gestures (which will be used in the first half of the project) and one for dynamic gestures (which ideally will be used in the second half of the project).

5.1 Static Datasets

- Synthetic ASL Finger-Spelling Dataset by Lexset [16]

5.2 Dynamic Datasets

- IPN Hand Dataset (https://gibranbenitez.github.io/IPN_Hand/)
- JESTER dataset (<https://www.kaggle.com/datasets/toxicmender/20bn-jester>)
- DVS128 dataset (https://coldpress.ai/product_page/1680201883079x468828458487031500)

6 Evaluation & Discussion

The models will be evaluated on the intra-class and inter-class cosine-similarities, T-SNE projection, and prediction accuracies based on clustering.

7 Future Plans

In this project, we plan to integrate the keypoint stream into our model in the following month, with the keypoints being extracted using efficient keypoint extraction models such as MediaPipe [9].

In the future, we plan to explore on:

- Semi-Hard Mining for Triplet Loss

- Using InfoNCE loss
- Generate dense fake data using keypoints to enhance the model’s understanding on what is considered ”similar” gestures.

8 Proposed timelines

September • Study foundation papers related to the target problem and decide the basic directions of the project.

October • Study papers related to the architectures and methodologies that could be used to improve the efficiency.

- Train models on smaller datasets to test if the concept is feasible.
- Focuses mainly on static gestures.
- Modify the directions of the project.

November • Continue training models on datasets to test if the concept is feasible.

- Perform more detailed analysis on the methods and their accuracies and efficiencies.
- Explore the weaknesses of the methods.

December – January • Explore on dynamic gestures.

February – March • Analysis on the accuracies and efficiencies of the methods.

- Explore the possibilities of auto-annotation of sign language videos for previously unseen hand gestures.

References

- [1] Y. Chen, R. Zuo, F. Wei, Y. Wu, S. Liu, and B. Mak, *Two-stream network for sign language recognition and translation*, 2023. arXiv: 2211.01367 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2211.01367>.
- [2] O. Yusuf, M. Habib, and M. Moustafa, *Real-time hand gesture recognition: Integrating skeleton-based data fusion and multi-stream cnn*, 2024. arXiv: 2406.15003 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2406.15003>.
- [3] M. Tan and Q. V. Le, *Efficientnet: Rethinking model scaling for convolutional neural networks*, 2020. arXiv: 1905.11946 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/1905.11946>.
- [4] G. R. Koch, “Siamese neural networks for one-shot image recognition,” 2015. [Online]. Available: <https://api.semanticscholar.org/CorpusID:13874643>.
- [5] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, *A simple framework for contrastive learning of visual representations*, 2020. arXiv: 2002.05709 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2002.05709>.
- [6] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, *Momentum contrast for unsupervised visual representation learning*, 2020. arXiv: 1911.05722 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/1911.05722>.
- [7] J.-B. Grill, F. Strub, F. Altché, *et al.*, *Bootstrap your own latent: A new approach to self-supervised learning*, 2020. arXiv: 2006.07733 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2006.07733>.

- [8] Z. Yang, A. Zeng, C. Yuan, and Y. Li, *Effective whole-body pose estimation with two-stages distillation*, 2023. arXiv: 2307.15880 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2307.15880>.
- [9] C. Lugaresi, J. Tang, H. Nash, *et al.*, *Mediapipe: A framework for building perception pipelines*, 2019. arXiv: 1906.08172 [cs.DC]. [Online]. Available: <https://arxiv.org/abs/1906.08172>.
- [10] G. Sung, K. Sokal, E. Uboweja, *et al.*, *On-device real-time hand gesture recognition*, 2021. arXiv: 2111.00038 [cs.CV]. [Online]. Available: <https://arxiv.org/abs/2111.00038>.
- [11] W. Zhong, Y. Zhang, P. Fu, W. Xiong, and M. Zhang, *A spatio-temporal graph convolutional network for gesture recognition from high-density electromyography*, 2023. arXiv: 2312.00553 [cs.HC]. [Online]. Available: <https://arxiv.org/abs/2312.00553>.
- [12] J. Ahn, Y. Jang, and J. S. Chung, “Slowfast network for continuous sign language recognition,” in *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024, pp. 3920–3924. DOI: 10.1109/ICASSP48485.2024.10445841.
- [13] A. van den Oord, Y. Li, and O. Vinyals, *Representation learning with contrastive predictive coding*, 2019. arXiv: 1807.03748 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/1807.03748>.
- [14] I. Dave, R. Gupta, M. N. Rizve, and M. Shah, “Tclr: Temporal contrastive learning for video representation,” *Computer Vision and Image Understanding*, vol. 219, p. 103406, Jun. 2022, ISSN: 1077-3142. DOI: 10.1016/j.cviu.2022.103406. [Online]. Available: <http://dx.doi.org/10.1016/j.cviu.2022.103406>.
- [15] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence*, ser. IJCAI-2018, International Joint Conferences on Artificial Intelligence Organization, Jul. 2018, pp. 3634–3640. DOI: 10.24963/ijcai.2018/505. [Online]. Available: <http://dx.doi.org/10.24963/ijcai.2018/505>.
- [16] Lexset, *Synthetic asl alphabet*, Jun. 2022. [Online]. Available: <https://www.kaggle.com/datasets/lexset/synthetic-asl-alphabet>.